

Decode Labs Data Analytics Internship

Week 4 – Project 4: Data Visualization & Storytelling

Submitted By: Ifrah Aftab

Internship: Decode Labs Data Analytics Internship

Project Title: Data Visualization and Storytelling Using Microsoft Excel

Objective

The objective of this project was to transform business data into meaningful visualizations using Microsoft Excel. Different chart types were used to identify trends, compare performance, analyze relationships, and communicate business insights effectively. The project focused on applying data storytelling techniques to support data-driven decision-making.

Dataset Overview

The dataset used for this project was the cleaned e-commerce dataset prepared during Week 1 of the internship.

The dataset contains information related to:

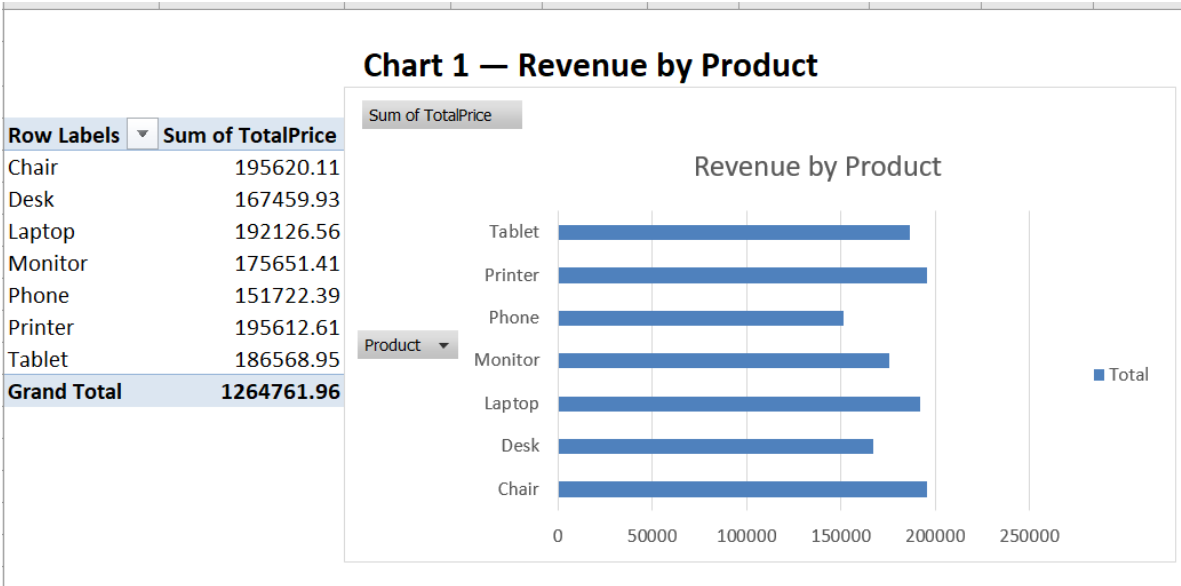
- Customer Orders
- Products
- Quantities Sold
- Unit Prices
- Total Revenue
- Payment Methods
- Order Status
- Referral Sources

Total Records Analyzed: 1,200 Orders

Visualization 1 – Revenue by Product

Chart Type: Horizontal Bar Chart

Purpose: To compare total revenue generated by each product category.

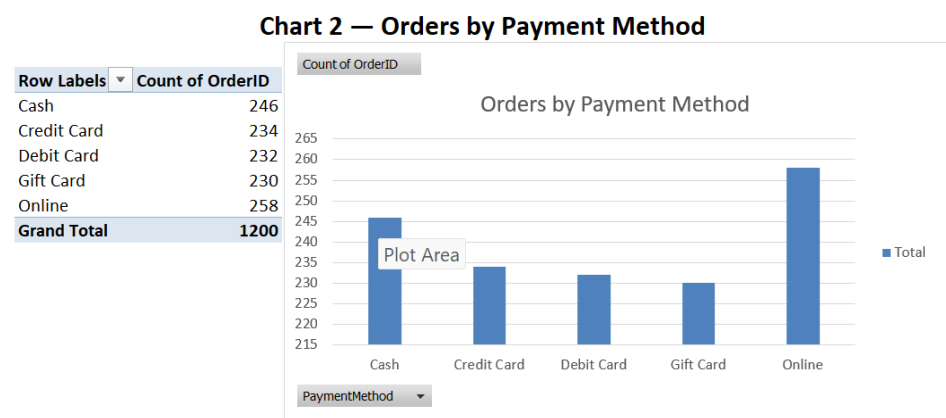


Insight: Chair products generated the highest revenue, contributing approximately **195,620** in sales. Printers followed very closely, indicating that both product categories are major contributors to overall business performance.

Visualization 2 – Orders by Payment Method

Chart Type: Column Chart

Purpose: To compare customer usage of different payment methods.

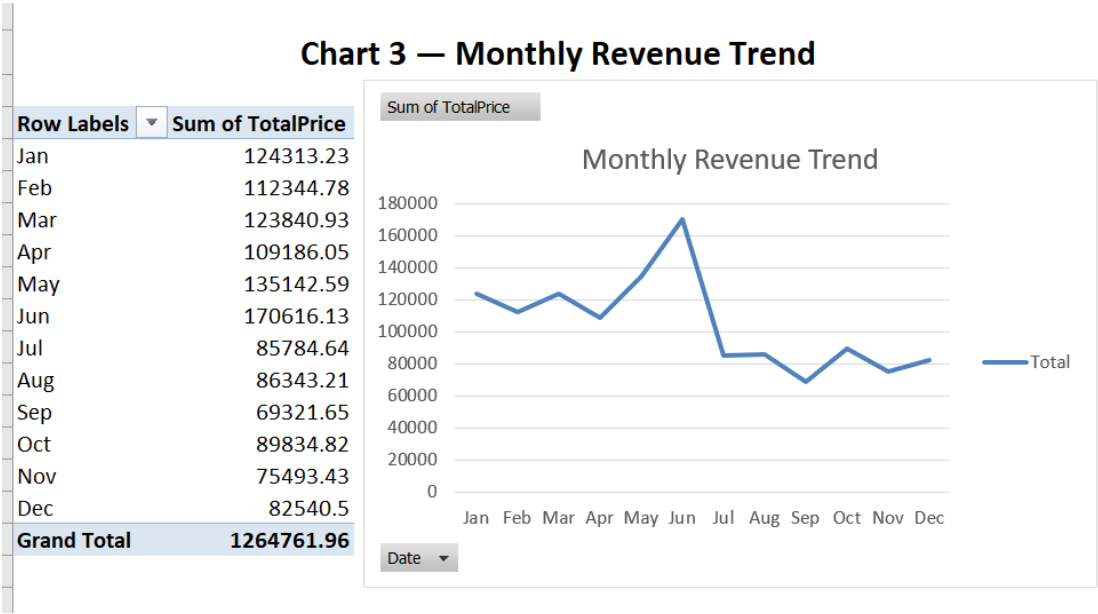


Insight: Online payments recorded the highest number of transactions with approximately **258 orders**, demonstrating a strong customer preference for digital payment options over other available methods.

Visualization 3 – Monthly Revenue Trend

Chart Type: Line Chart

Purpose: To analyze revenue performance over time and identify trends.



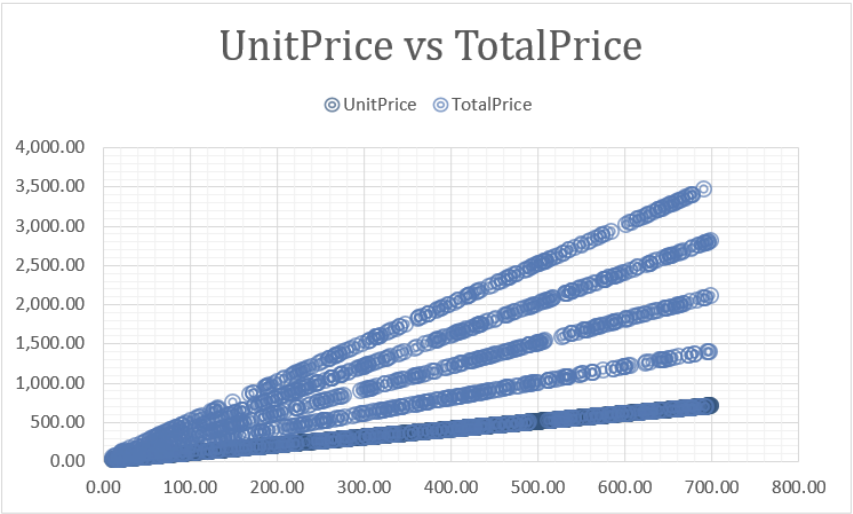
Insight: Revenue levels fluctuated throughout the observed period, reflecting variations in customer purchasing activity. Trend analysis provides valuable insights into periods of stronger and weaker business performance.

Visualization 4 – Relationship Between Unit Price and Total Price

Chart Type: Scatter Plot

Purpose: To examine the relationship between product pricing and total transaction value.

Chart 4- UnitPrice vs TotalPrice

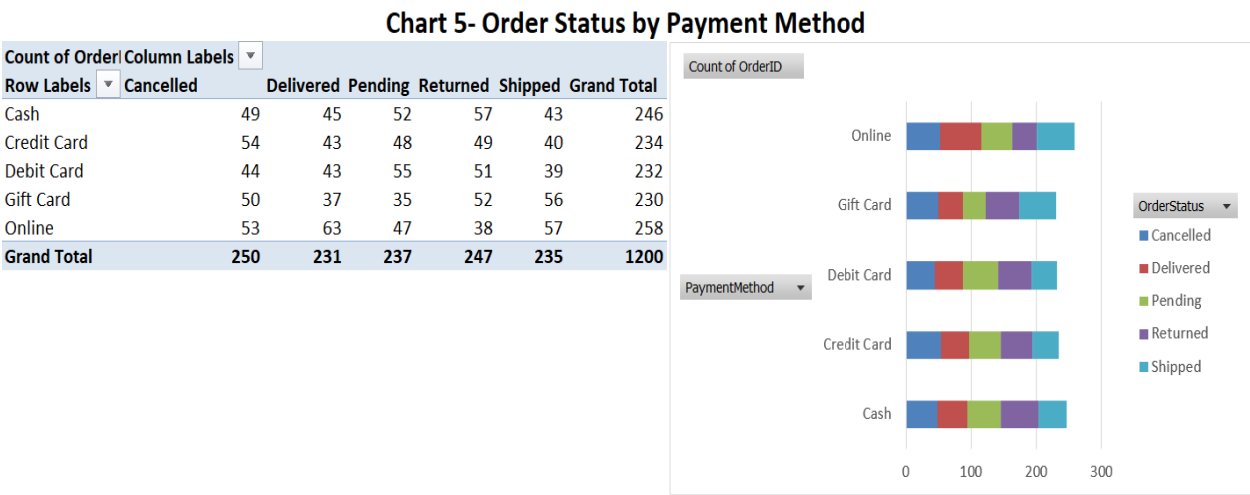


Insight: The scatter plot demonstrates a positive relationship between unit price and total order value. Higher-priced products generally contribute to larger transaction amounts, highlighting the impact of pricing on revenue generation.

Visualization 5 – Order Status by Payment Method

Chart Type: Stacked Bar Chart

Purpose: To compare order status distributions across different payment methods.

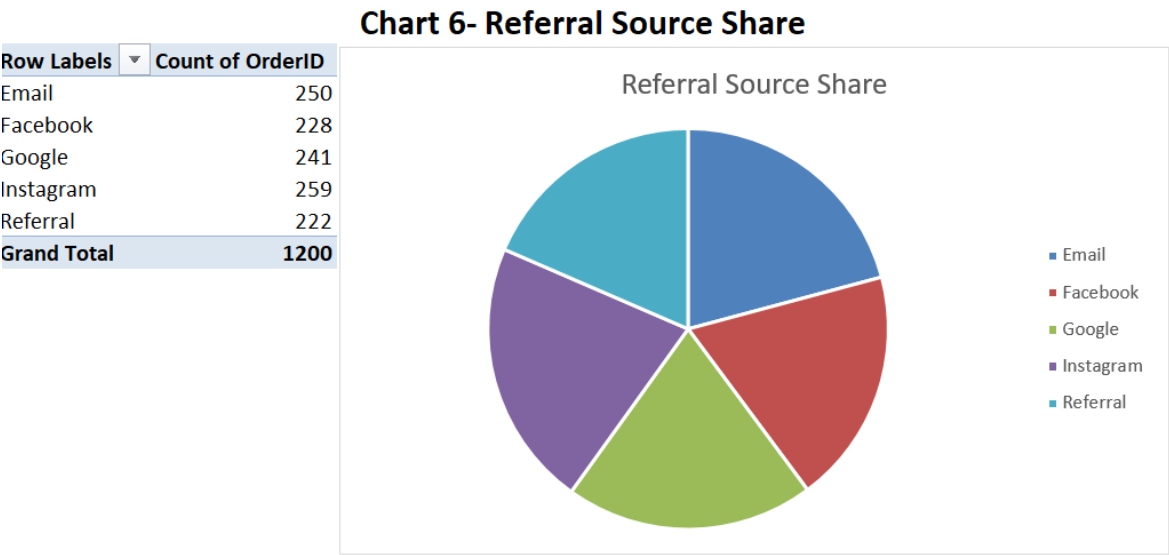


Insight: Most transactions progressed successfully through the fulfillment process across all payment methods. The visualization highlights differences in order outcomes and provides insight into operational performance and transaction management.

Visualization 6 – Referral Source Distribution

Chart Type: Pie Chart

Purpose: To analyze the contribution of different referral sources to total customer orders.



Insight: The pie chart illustrates the distribution of customer orders across referral sources. Certain referral channels contributed a larger share of customer acquisitions, indicating that some marketing sources are more effective than others in driving traffic and sales.

Key Findings

- Chair products generated the highest revenue among all product categories.
 - Online payments were the most frequently used payment method.
 - Revenue patterns fluctuated over time, highlighting changes in sales performance.
 - Higher unit prices generally resulted in higher transaction values.
 - Most customer orders were processed successfully regardless of payment method.
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Conclusion

This project demonstrated the importance of data visualization in transforming raw business data into actionable insights. By utilizing multiple chart types, including bar charts, column charts, line charts, scatter plots, and stacked bar charts, complex datasets were presented in a clear and meaningful manner. The visualizations enabled the identification of key trends, customer behavior patterns, revenue drivers, and operational insights. Overall, the project strengthened practical skills in Microsoft Excel, data visualization, and business storytelling.